

Mind in Motion: Investigating how Walking in Physical and Virtual Environments Influence Programming Problem Solving

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Abstract

Programming is an activity that often requires one to plan and reflect on their problem solving process. Recent work suggests that physical movements can help during problem solving. However, little is known about the effects of movement on programming problem solving, and whether movement conducted in virtual environments share similar cognitive benefits to physical walking. This study explores differences between participants taking a physical walk (PW), and a walk in a virtual reality environment (VRW), on their ability to solve programming problems. We conducted a pilot study with 12 participants to evaluate differences in task completion time and code correctness in PW and VRW conditions. We found that physical walking led to higher accuracy in solving programming problems compared to walking in a VR environment. However, task completion time increased in both conditions relative to the baseline. We discuss implications and future research directions in light of these findings.

CCS Concepts

• **Human-centered computing** → *Empirical studies in HCI*; • **Social and professional topics** → *Computing education*.

Keywords

Metacognition, Programming, Embodied Cognition, Virtual Reality, Walking

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1 Introduction

Programming is not only a technical task; it requires careful thinking and awareness of one's own problem solving process. At its core, programming requires planning, keeping track of progress, checking intermediate results, and fixing logical errors. These activities rely on programmers metacognitive skills, which allow them to remain aware of what they understand, what they do not, and how they might re-strategize to move toward a solution [21]. When metacognition fails, programmers often fall into “cognitive lock,” where they become fixated on one way of thinking and struggle to consider alternatives. A growing body of research in embodied cognition [7] suggests that cognitive processes are not isolated within the mind but are influenced by the physical actions and environment around the individual. Movement, especially simple and steady actions like walking, has been shown to help people think more flexibly, come up with new ideas, and reorganize their thoughts [16]. This leads to an interesting question: Can short walking interventions prior to a programming task support more effective metacognitive monitoring and, consequently, improve task performance? While physical walking has been studied extensively for its cognitive benefits [23], research on virtual locomotion especially in a simulated virtual environment is still emerging [22]. Virtual reality (VR) creates immersive environments that change how the body and the surroundings interact, which can influence how people think. Because moving in VR requires adjusting perception, coordinating movements, and sometimes reorienting in space, it may place a different kind of mental demand on users than normal walking. It is still not clear whether walking in VR can provide the same benefits as real walking, improve them, or reduce them, and this remains an open question for research. Our work addresses this gap by examining how Physical Walking (PW) and Virtual Reality Walking (VRW) differentially affect programming performance and accuracy. Our findings from a pilot study show that short walking breaks may influence how students approach programming, with participants performing better (i.e. better code quality) after physical walking compared to walking in VR. This study contributes to ongoing discussions about how immersive



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technologies affect thinking, how VR interventions should be designed based on evidence, and how movement-based strategies can support students learning computer science.

2 Related Work

2.1 Metacognition in Programming Education

Metacognition commonly described as “thinking about thinking” refers to learners’ awareness and regulation of their own cognitive processes, including planning, monitoring, and evaluating strategies and outcomes [9, 15, 24]. In the context of programming education, metacognitive skills guide how students approach a new problem, select an algorithm, anticipate failure modes, monitor intermediate program behavior, and evaluate whether their solution meets the specification [13, 19]. Metacognition is central to programming because programmers must plan, monitor, and revise their thinking, especially during debugging, where they test hypotheses and update their understanding of the program [12, 20]. Beck et al. present a machine learning approach to provide real-time feedback on programming metacognition by analyzing students’ source code comments [3]. Pechorina et al. proposed *Metacodenition*, a programming environment that integrates metacognitive scaffolding and improves both novice students’ coding performance and their acceptance of the tool [18]. Li et al. found that teaching metacognitive strategies helps non-CS students stay engaged, manage negative emotions, and avoid unproductive behaviors in an introductory CS course [11]. Recent work also shows that metacognitive strategy use varies with student resilience, with high-resilience students adopting more planning based approaches, while low-resilience students following more reactive problem-solving patterns [4].

2.2 Physical Activity in Natural and Virtual Settings

Previous studies have shown that nature walks can improve mood, attention, and problem solving [10], and that virtual green environments can also reduce stress and improve mood [6]. Pasanen et al. found that nature walks improve mood and restoration and can also support sustained attention [17]. Bailey et al. showed that both indoor and outdoor walking improve cognitive performance, with outdoor walking leading to greater mental restoration measured by behavioral tests and EEG [1]. Similarly, Mavrantza et al. found that physical activity in natural environments provides stronger psychological benefits than indoor exercise [14]. Their results suggest that these effects may be linked to changes in brain activity and connectivity. Green exercise, which combines physical activity with nature exposure, provides added benefits over physical activity alone. Although immersive virtual environments have been proposed to simulate such experiences, it remains unclear whether they can produce effects comparable to real natural settings. Prior work suggests that limitations such as reduced sensory richness, postural instability, and cybersickness can weaken immersion and effectiveness in immersive virtual environments [5]. At the same time, simply being outdoors can improve cognitive performance even without walking, highlighting the role of environmental context [2]. Virtual green exercise can produce positive effects, but these benefits remain weaker than those of real outdoor activity

[14]. However, the specific effects of walking itself versus virtual experiences on complex problem-solving tasks such as programming remain largely unexplored.

3 Research Design

Our work aims to investigate how problem-solving performance is influenced by two different walking conditions: *Physical Walking (PW)* and *Virtual Reality Walking (VRW)*. The experiment examines whether walking in a physical environment versus an immersive virtual environment differentially affects participants’ time to solve a problem and the quality of their code solution. Participant experiences conditions in a counterbalanced order to control for order effects. The entire study is structured into three main sessions: Baseline, Intervention-1, and Intervention-2 separated by a 6 hours break.

3.1 Research Questions and Hypotheses

In this work we aim to address the following research questions:

- **RQ 1:** How does problem-solving time differ between baseline, physical walking, and virtual reality walking conditions?
- **RQ 2:** How do physical walking and virtual reality walking affect the quality of programming solutions compared to the baseline?

3.2 Participants

The sample size was 12 students (Male=8, Female=4) enrolled at a South-Asian Liberal Arts university. They were selected using purposive sampling to ensure appropriate background knowledge and suitability for the tasks. Before participating in the study, we ensured that all participants (1) had completed a basic programming course, (2) were comfortable walking for 5–10 minutes, and (3) were comfortable using a VR headset for 5–10 minutes. All participants were physically present on the university campus at the time of the experiment.

3.3 Experimental Setup

The VR condition was conducted using a head-mounted display (HMD) with 6-DOF tracking, developed in Unity using ProBuilder for custom 3D environments. The physical walking condition took place in a designated open space with similar spatial dimensions. Both settings aimed to simulate a comparable walking experience, maintaining a consistent walking duration (Fig. 1). Participants interacted with a virtual environment that mimicked natural locomotion and allowed spatial exploration before solving programming problems on a desktop computer. All problems were of similar difficulty level, ensuring equivalence across sessions. The programming problems used in this study were intentionally designed to be at a basic level, focusing on fundamental logic and control structures rather than advanced concepts.

3.4 Procedure

The study followed a controlled, repeated-measures design consisting of four stages: (1) a baseline session, (2) the first intervention session, (3) a break period, and (4) the second intervention session,



Figure 1: Glimpses of the study showing a student solving the problem (left), student doing a simulated walk in VR (center), and a student doing a physical walk (right)

as illustrated in Fig. 2. Each session included a reading phase, a walking (or rest) phase, and a problem-solving phase. In the baseline session, participants read the first programming problem (2 minutes), took a short rest (they were stationary, sitting at their desk), and then solved the problem on a computer (10 minutes) to establish baseline measures. In the first intervention session, participants read the second problem (2 minutes), performed Physical Walking (PW) for 5 minutes and thought about the problem, and then attempted the problem on a computer (10 minutes). The two intervention sessions were separated by a 6-hour break to reduce carry-over effects. In the second intervention session, participants read the third problem (2 minutes), performed the Virtual Reality Walk (VRW) for 5 minutes and thought about the problem, and then attempted the problem on a computer (10 minutes).

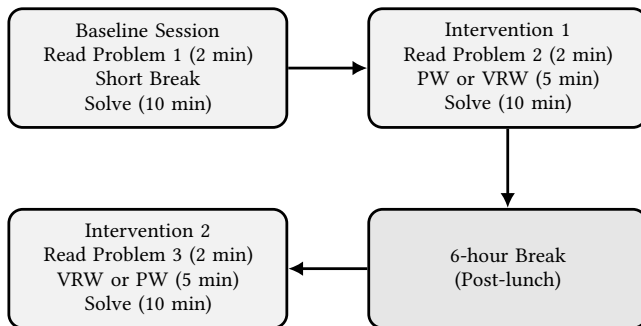


Figure 2: Overview of the study procedure. PW = Physical Walking; VRW = Virtual Reality Walking.

4 Data Analysis and Findings

The evaluation strategy for this study was designed to capture objective programming performance associated with the physical walk and virtual walk interventions. As the present work constitutes a pilot study conducted with 12 participants, the evaluation focuses on descriptive and exploratory analysis rather than inferential statistical testing. The main goal was to check whether the study design is appropriate, and to look for early patterns that could guide a larger future study.

4.1 Quantitative Measures

Two primary quantitative measures were employed to evaluate programming performance across experimental conditions. The first measure was *task completion time*, recorded in seconds, which represents the duration required by a participant to complete each programming task. Task time serves as an indicator of cognitive efficiency and problem-solving fluency, particularly in tasks that require sustained attention, iterative debugging, and strategic reasoning. To ensure consistency, all participants were allotted a maximum of ten minutes (600 seconds) per task, and time measurements were recorded uniformly across the baseline, Physical Walking (PW), and Virtual Reality Walking (VRW) conditions. The second quantitative measure was a *Code Quality Rubric* proposed in this study (see Table 1). Submissions were evaluated using a five-point scale ranging from failed attempts (0) to fully correct solutions (5). Since the participants were second-year computer science students, the assessment focused primarily on the correctness and completeness of the underlying logic rather than code style or advanced optimizations. The rubric considers functional correctness, handling of edge cases, and algorithmic correctness, allowing a more precise comparison between partially correct and fully correct solutions. Cohen’s kappa (κ) was calculated to assess inter-rater reliability between two observers to calculate the code quality score across 36 paired ratings (12 participants who solved 1 problem in the baseline and 1 problem in each intervention). The analysis revealed a $\kappa = 0.89$, indicating almost perfect agreement [8].

Table 1: Code Quality Scoring Rubric

Score	Category	Definition
0	Completely incorrect	No code submitted, or the submitted code is unrelated to the problem.
1	Serious attempt, but non-functional	Student attempted the task, but the logic is fundamentally incorrect; only a trivial part is implemented.
2	Mostly incorrect	Student attempted the task, but the logic works only for trivial or special cases.
3	Partially correct solution	Student attempted the task; the core logic is correct, but the implementation is incomplete or flawed and misses important conditions.
4	Almost correct solution	Student attempted the task; the logic is mostly correct but contains a minor bug or fails on rare edge cases.
5	Fully correct solution	Student attempted the task; the logic is completely correct and produces the correct output for all test cases.

Table 2: Mean task completion time and code quality across conditions, with change relative to baseline.

Condition	Mean Time (seconds)	ΔT_c (seconds)	Mean Score	ΔS_c
Baseline	565.0	–	1.5	–
Physical Walking (PW)	587.5	+22.5	3.17	+1.667
Virtual Reality Walking (VRW)	582.55	+17.5	2.33	+0.833

4.2 Findings

With respect to RQ 1 (differences in problem-solving time), both physical walking (PW) and virtual reality walking (VRW) took longer task completion times compared to the baseline condition. This can suggest that after both interventions, participants spent more time reflecting and planning before or during coding. The increase in time was larger after physical walking than after VR walking, which can indicate that physical walking encouraged a more deliberate and careful problem-solving approach. Regarding RQ 2 (differences in solution quality), physical walking led to a clear improvement in code quality scores compared to baseline. In contrast, virtual reality walking showed a smaller and less consistent improvement, and in some cases a slight decline in performance. Table 2 summarizes the mean task completion time (T) and mean code quality score (S) across the three experimental conditions. To quantify the effect of each intervention relative to baseline, we define the change in performance as:

$$\Delta T_c = T_c - T_{\text{baseline}}, \quad \Delta S_c = S_c - S_{\text{baseline}}, \quad (1)$$

where $c \in \{\text{baseline}, \text{PW}, \text{VRW}\}$.

Using these definitions, PW yields $\Delta T_{\text{PW}} = +22.5$ s and $\Delta S_{\text{PW}} = +1.667$, whereas VRW yields $\Delta T_{\text{VRW}} = +17.5$ s and $\Delta S_{\text{VRW}} = +0.833$. Thus, although both interventions increased task completion time ($\Delta T > 0$), both physical walking and virtual walking improved solution quality, but virtual reality walking was associated with a slight decline in code quality score. This seems to suggest that the immersive nature of VR may introduce additional cognitive load that partially offsets its benefits.

5 Discussion

For RQ 1, both physical walking and VR walking increased problem-solving time compared to baseline, with physical walking leading to a slightly larger increase, which might suggest more careful problem-solving. For RQ 2, physical walking improved code quality, while VR walking showed less consistent effects, sometimes even slightly reducing solution quality. Overall, the results suggest that short walking breaks can influence how students approach programming problems, but the effect depends strongly on the type of walking. Physical walking appears to help participants think more clearly and flexibly, consistent with earlier work showing that real walking and nature exposure can improve attention and cognitive state [1, 10, 14, 17]. One possible reason is that physical walking helps people break out of unproductive thought patterns and return to the task with a fresh mind. In contrast, VR walking did not show the same benefit and was sometimes linked to a small drop in solution quality. This matches earlier studies showing that immersive virtual environments can require extra mental effort, such as adjusting to body movement, limited sensory feedback, or

mild discomfort, which can reduce their relaxing and refreshing effect [5, 14]. Importantly, these findings do not imply that VR is unsuitable; rather they highlight that VR-based interventions must be carefully designed. Factors such as navigation control, adaptation to the virtual space and user comfort likely influenced participants' cognitive state after the VR walk.

In this study, we have considered code quality and completion time as indirect indicators of metacognitive processes. While longer completion times and higher-quality solutions may indicate planning, monitoring, and evaluation behaviors, more direct evidence would require think-aloud protocols or other process-tracing approaches (e.g., screen recordings, keystroke logs, or reflective questionnaires). Future studies can incorporate integrated process-tracing measures to more directly capture planning, monitoring, and evaluation behaviors during programming tasks.

A key limitation of this study is the small sample size, due to which the findings cannot be generalized to larger populations. Additionally, the programming tasks were intentionally basic, limiting insight into how embodied interventions may influence more complex problem-solving scenarios. Although fixed timing ensured experimental control, it does not fully reflect authentic programming contexts, where walking breaks are typically self-initiated. Examining context-driven interventions remains an important direction for further investigation. However, the patterns observed are meaningful and suggest that the study design and evaluation methods are suitable for capturing changes in both problem-solving behavior and code quality. We also acknowledge cognitive load, novelty effects, and potential VR discomfort as possible confounds that may have influenced the results. A study with more participants and additional measures of cognitive load, will be necessary to confirm these trends and better understand how physical and virtual movement differently support programming and metacognitive processes.

6 Conclusion and Future Work

This study explored whether short walking-based breaks, either in the physical world or in virtual reality, can influence how students approach programming problems. Although the study was small and exploratory, the results suggest that physical walking may help students return to programming tasks with clearer thinking and slightly better solution quality, while virtual walking may introduce additional mental load that reduces this benefit. These findings support the idea that not all forms of embodied or immersive activity affect cognition in the same way, and that the design of such interventions matters.

As part of future work, we intend to incorporate integrated process-level measures to enable more direct and reliable assessment of metacognitive planning, monitoring, and evaluation behaviors during programming tasks. A larger and more diverse sample, together with more complex and ecologically valid programming scenarios, will be necessary to strengthen the generalizability of the findings. Examining self-initiated, context-driven walking breaks would better reflect authentic programming practice, while additional measures of cognitive load and VR discomfort would help clarify potential confounds. A particularly promising direction is the exploration of mixed reality systems in which participants wear see-through or mixed reality glasses during physical walking. In such setups, the programming problem could remain visible while moving, enabling continuous engagement with the task and potentially combining the cognitive benefits of real physical movement with sustained problem-focused reflection.

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